

Sarthak Sinha

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Software Engineer | Full-Stack & Backend Developer | AI-Integrated Applications

Professional Summary

Software Engineer experienced in building and shipping production web applications, backend systems, and AI-integrated platforms using Node.js, React, and MongoDB. Strong foundation in REST API design, data structures & algorithms, and applied machine learning, with a track record of delivering full-stack projects end-to-end.

Key Highlights

- Delivered **3+ full-stack applications** (video hosting platform, agentic RAG chatbot, AI e-learning platform) with secure auth, database design, and real-time/AI features.
- Software Engineer at Accenture**, contributing to live web releases across the full development-to-deployment cycle.
- Secured **2 hackathon podium finishes** (GENEVOE 2024, HACKRUSH 1.0) building applied AI/ML solutions under time constraints.

Experience

Software Engineer

July 2026 – Present

Accenture

- Built and tested responsive web pages and components with a cross-functional team, supporting deployment and post-release debugging across multiple release cycles.
- Applied industry-standard coding and design practices on live production web projects, improving consistency and maintainability of delivered code.
- Used Git-based workflows and participated in regular code reviews, contributing to improved code quality and reduced defect rates.

Projects

Kinescope – Full-Stack Video Platform with AI Agents

React, Node.js, Express.js, MongoDB, LangGraph, Gemini

API, Cloudinary

[Live Demo](#)

- Built a full-stack YouTube-style platform – React, Node/Express, MongoDB – with Cloudinary video upload, comments, subscriptions, and playlists; deployed on Render with GitHub CI/CD.
- Designed **2 LangGraph AI agents** on Gemini: a RAG-based recommendation engine grounded in a user's real watch/like history, and a hallucination-safe video-chat assistant that answers only from retrieved video metadata.

Adaptive RAG Chatbot (Agentic AI, Full-Stack)

Python, FastAPI, LangGraph, LangChain, Groq API, FAISS,

MongoDB, Streamlit

[Live Demo](#)

- Built an agentic RAG chatbot using LangGraph that classifies each query with an LLM and dynamically routes it to document retrieval, general knowledge, or live web search (Tavily API).
- Designed a self-correcting retrieval pipeline with LLM-based relevance grading and automatic query rewriting on low-relevance results, backed by a FAISS vector store and Groq-hosted LLMs.
- Deployed a full-stack production app (FastAPI backend on Render, Streamlit frontend on Streamlit Cloud) with persistent, session-based chat history in MongoDB Atlas.

LearnMind – AI-Assisted E-Learning Platform (Backend)

React, Zustand, Node.js, Express, MongoDB, Socket.IO,

Gemini API

- Built a full-stack, AI-assisted e-learning platform with role-based authentication; integrated the Gemini API for an adaptive assessment engine generating personalized exams and an AI tutoring chatbot.
- Designed student diagnostic dashboards for real-time performance insights and added Socket.IO notifications to boost engagement.

Education

Bachelor of Technology in Computer Science and Engineering

June 2022 – May 2026

SRM Institute of Science and Technology

CGPA: 8.63/10

Technical Skills

Languages: C++, Python, SQL, PL/SQL, JavaScript, HTML, CSS

Backend & APIs: Node.js, Express.js, FastAPI, REST API Development, Socket.IO, JWT Authentication, Role-Based Access Control, Middleware Design

Databases: MongoDB, Mongoose, MongoDB Atlas, SQL, Oracle PL/SQL, Oracle APEX

AI / Machine Learning: Gemini API, Groq API, LangChain, LangGraph, AI Chatbot Integration, RAG (Retrieval-Augmented Generation), Agentic AI, Embeddings, Vector Databases (FAISS), Prompt Engineering

Tools & Platforms: Git, GitHub, Postman, Jupyter Notebook, Cloudinary, Zustand, Streamlit, Render, Docker

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming, API Testing, Data Analytics, Cloud Fundamentals

Achievements

- 1st Runner-Up, HACKRUSH 1.0 (EdTech Track)** – Built a chatbot for automated research paper scraping and PDF summarization. 2024

Certifications

- Alteryx Designer Core